

Caffeine Impairs Myocardial Blood Flow Response to Physical Exercise in Patients with Coronary Artery Disease as well as in Age-Matched Controls

Background Caffeine is one of the most widely consumed pharmacologically active substances. Its acute effect on myocardial blood flow is widely unknown. Our aim was to assess the acute effect of caffeine in a dose corresponding to two cups of coffee on myocardial blood flow (MBF) in coronary artery disease (CAD). Methodology/Principal Findings MBF was measured with ^{15}O -labelled H_2O and Positron Emission Tomography (PET) at rest and after supine bicycle exercise in controls ($n = 15$, mean age 58 ± 13 years) and in CAD patients ($n = 15$, mean age 61 ± 9 years). In the latter, regional MBF was assessed in segments subtended by stenotic and remote coronary arteries. All measurements were repeated fifty minutes after oral caffeine ingestion (200 mg). Myocardial perfusion reserve (MPR) was calculated as ratio of MBF during bicycle stress divided by MBF at rest. Resting MBF was not affected by caffeine in both groups. Exercise-induced MBF response decreased significantly after caffeine in controls (2.26 ± 0.56 vs. 2.02 ± 0.56 , $P < 0.005$), remote (2.40 ± 0.70 vs. 1.78 ± 0.46 , $P < 0.001$) and in stenotic segments (1.90 ± 0.41 vs. 1.38 ± 0.30 , $P < 0.001$). Caffeine decreased MPR significantly by 14% in controls ($P < 0.05$ vs. baseline). In CAD patients MPR decreased by 18% ($P < 0.05$ vs. baseline) in remote and by 25% in stenotic segments ($P < 0.01$ vs. baseline). Conclusions We conclude that caffeine impairs exercise-induced hyperaemic MBF response in patients with CAD to a greater degree than age-matched controls.